

Course Code : ECT 359-3

MQNR/RS – 23 / 1029

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

BIOMEDICAL ELECTRONICS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Draw neat and clean diagrams, wherever necessary.
- (3) Assume suitable data wherever necessary.

1. (A) Describe the various origins of bioelectric signals. Draw a typical cell potential waveform, label it properly and explain the phenomena of depolarization and repolarization. 5(CO1,2)
- (B) List the any five human physiological systems. Elaborate the working of human heart with respect to cardiovascular (circulatory) system. 5(CO1)
2. (A) Summarize and elaborate the various types of electrodes used in bio-potential measurement with their applications. 5(CO2)
- (B) What do you mean by transducer ? List out the different types of transducer used in measurement of temperature in medical field. Explain the principle of thermocouple. What is the most common thermocouple used for body temperature measurement ? 5(CO2)
3. (A) Elaborate on the differences between ECG, EEG, EMG and ERG Bioelectric signals with respect to their primary signal characteristics (frequency and amplitude ranges) types of electrode used in them and their applications medical diagnosis. 5(CO1,2)
- (B) What do you mean by bio-telemetry ? Which type of modulation system is used in a multichannel bio telemetry system ? Draw and briefly elaborate the single channel telemetry system suitable for transmission of an ECG. 5(CO1,4)

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4. (A) What is the principle of working of 'Pulse Oximeter' ? What type of transducer is used for pulse oximetry ? Explain the signal processing arrangement in a Pulse Oximeter. 5(CO3)
- (B) Describe the various scanning techniques used in computed tomography. Explain how the progressive developments in scanning techniques have helped to reduce the scanning time. 5(CO3)
5. (A) What are the general parameters monitored by Bedside Monitors ? Explain the generalized block diagram of a typical bedside monitor. 5(CO4)
- (B) What is the need of 'Ambulatory Monitoring Systems' ? Explain the sub-systems for data recording and analysis in an ambulatory monitoring system with an example. 5(CO4)
6. (i) Distinguish between gross shock and micro current shock. Briefly elaborate an effect of various levels of current on the human body.
- (ii) Draw the relationship between frequency and maximum permissible leakage currents through :
- (a) The body and
- (b) The heart and explain. 10(CO5)

